

Bayesian Game Theory

Definition

Bayesian Game Theory is an extension of game theory that studies situations where **players do not have complete information about each other**. Instead of knowing everything, players have **beliefs (probabilities)** about the unknown characteristics (called *types*) of other players and make decisions based on those beliefs.

It was introduced by **John Harsanyi to analyze games with incomplete information**.

The theory uses **Bayes' Theorem** to update beliefs when new information becomes available.

Bayes' Theorem:

$$P(A | B) = \frac{P(B | A) \times P(A)}{P(B)}$$

Meaning:

- Start with an initial belief (prior probability).
- Observe new information.
- Update your belief (posterior probability).

Key Concepts:

1. Incomplete Information
2. Types
3. Beliefs
4. Strategy
5. Payoffs

1. Incomplete Information

Players do **not know everything**.

Unknown information may include:

- Opponent's skill
 - Costs
 - Intentions
 - Resources
 - Preferences
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2. Types

A **type** describes a player's private information.

Examples:

- Honest seller
- Fraudulent seller
- Strong company
- Weak company
- Experienced bidder
- Beginner bidder

Each player knows **their own type**, but not necessarily the types of others.

3. Beliefs

Players assign probabilities to possible types.

Example:

A buyer believes:

- Honest seller = 80%
- Fraudulent seller = 20%

The buyer chooses an action based on these probabilities.

4. Strategy

A strategy specifies what a player will do **for every possible type** they might be.

Example:

If the seller is:

- Honest → Offer warranty.
 - Dishonest → Hide product defects.
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5. Payoffs

The payoff depends on:

- The chosen strategies.
- The players' types.
- The resulting outcomes.

Components of a Bayesian Game

A Bayesian game typically includes:

- Players
- Types
- Actions (strategies)
- Beliefs (probabilities)
- Payoffs

Example

Example 1: Buying a Used Car

Situation

A buyer wants to purchase a used car.

The seller can be:

- Honest
- Dishonest

The buyer does not know the seller's type.

The buyer believes:

- Honest = 70%
- Dishonest = 30%

Buy

Expected payoff:

$$0.70 \times 10 + 0.30 \times (-5) = 5.5$$

Don't Buy

Payoff = 0

Since **5.5 > 0**, the buyer decides to buy.

This is a Bayesian decision because the buyer uses beliefs about the seller's type.

Bayesian Nash Equilibrium

A **Bayesian Nash Equilibrium** occurs when:

Each player's strategy maximizes their expected payoff based on their beliefs about the other players' types, and no player can improve their expected payoff by changing their strategy alone.

Applications

Economics

- Auctions
- Pricing decisions
- Market competition

Finance

- Investment decisions under uncertain market conditions
- Credit risk assessment

Business

- Product launches when competitors' costs or strategies are unknown
- Negotiations and contract design

Artificial Intelligence

- Multi-agent systems
- Autonomous vehicles interacting under uncertainty
- Robot coordination

Cybersecurity

- Attackers and defenders making decisions with incomplete information

Politics

- Election campaigns
- International diplomacy
- Military strategy

Advantages

- Models real-world situations with incomplete information.
 - Incorporates uncertainty and beliefs into decision-making.
 - Widely used in economics, AI, auctions, and strategic planning.
 - Helps predict behavior when information is imperfect.
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Limitations

- Requires estimating probabilities, which may be inaccurate.
- Can become mathematically complex with many players or types.
- Assumes players update beliefs rationally.
- Real people may behave emotionally or irrationally.